12_Proposed_Solution

Proposed Solution

Integrated AICTE Approval Workflow Automation, Compliance Monitoring and Alert Management System

SIH Problem Statement: PS26262

1. Solution Overview

We propose an integrated, web-based AICTE approval workflow management system that streamlines the end-to-end processing of approval applications through configurable workflows, centralized pendency tracking, compliance monitoring, document expiry alerts, EVC scheduling, automated notifications, escalation management, and role-based dashboards.

The system will incorporate two additional innovation modules:

1. **Compliance Evidence Intelligence (CEI):** Connects regulatory requirements with their supporting documents, evaluates evidence completeness and validity, flags discrepancies, and explains what needs attention.
2. **Regulatory Change Impact Engine (RCIE):** Analyzes changes between regulatory rule versions and identifies applications, compliance requirements, documents, and pending tasks that may need reassessment.

These modules will work alongside a configurable workflow engine. The workflow engine coordinates tasks and permitted transitions, CEI provides evidence-level compliance insights, and RCIE helps administrators understand the potential operational impact of regulatory changes.

The system will support authorized human decision-making rather than autonomously granting, rejecting, or modifying statutory approvals.

2. Problem Addressed

Approval processes can involve multiple stakeholders, regulatory requirements, supporting documents, scrutiny stages, inspections or EVCs, deadlines, and post-approval obligations. Managing these activities requires visibility into the current status of applications, responsibility for pending actions, compliance evidence, and upcoming deadlines.

The proposed system addresses the following operational needs:

- Centralized tracking of applications and their current approval stages.
- Configurable routing of applications and tasks to authorized roles.
- Visibility into pending, overdue, and completed activities.
- Coordination and tracking of EVC schedules and related reports.
- Monitoring of undertakings, compliance obligations, and document expiry dates.
- Timely notifications and policy-based escalation of pending tasks.
- Consolidated reports for operational monitoring and decision support.
- Better traceability between regulatory requirements, supporting evidence, and administrative actions.
- Impact assessment when approved regulatory requirements or workflow rules change.

The actual capabilities of existing AICTE systems and the availability of integration interfaces will be validated before deployment. The proposed system will use authorized integrations where available and a controlled demonstration environment otherwise.

3. Core Proposed Solution

3.1 Configurable Workflow Automation

The platform will provide a configurable workflow engine to manage application processing across the relevant approval stages.

Key features:

- Configurable stages, dependencies, task assignments, and permitted transitions.
- Role-based task queues for institutions, scrutiny personnel, regional officers, compliance personnel, and approving authorities.
- Tracking of application status, pending actions, responsible roles, and task history.
- Configurable deadlines, reminders, and escalation rules.
- Support for exceptions, additional information requests, and resubmissions where applicable.
- A complete audit trail of status changes and user actions.

Workflow transitions will depend on configured business rules and required human actions. The system will not bypass mandatory scrutiny, statutory requirements, or competent-authority decisions.

3.2 Centralized Pendency and Operations Dashboard

A unified dashboard will provide an operational overview of applications and outstanding tasks.

It will display:

- Total applications by stage and status.
- Pending and overdue tasks.
- Applications awaiting documents, reports, or verification.
- Upcoming EVCs and related deadlines.
- Expiring documents and outstanding compliance obligations.
- Workload distribution and stage-wise processing trends.
- Escalations requiring attention.

Authorized users will see information appropriate to their roles and permissions. Institutions will be able to track their own applications and outstanding obligations, while authorized administrators will be able to view permitted cross-application summaries.

3.3 EVC Scheduling and Coordination

The system will support the scheduling and monitoring of Expert Visiting Committee activities, where applicable.

Features will include:

- EVC scheduling and assignment tracking.
- Recording of schedule details and authorized participants.
- Notifications for upcoming visits and pending preparation tasks.
- Tracking of EVC reports and report submission status.
- Visibility into applications awaiting EVC-related action.

Actual committee selection, scheduling constraints, and procedural exceptions will follow the applicable authorized rules.

3.4 Compliance and Document Expiry Monitoring

The platform will maintain records of relevant undertakings, compliance obligations, and supporting documents.

It will track:

- Document type and associated application or institution.
- Submission date and document version.
- Applicable validity or expiry date, where relevant.
- Verification status and verification history.
- Upcoming expiries and overdue obligations.
- Responsible users and outstanding actions.

Configurable notifications will alert relevant users before expiry and when obligations remain incomplete. Expiry or missing-document flags will inform authorized users; they will not automatically determine an institution's statutory eligibility.

3.5 Notifications, Escalations and Reporting

A centralized notification service will send configurable reminders and alerts through supported channels.

Examples include:

- An application remaining pending beyond its configured deadline.
- A required report not yet submitted.
- A document approaching its recorded expiry date.
- An institution needing to provide additional evidence.
- An assigned task awaiting verification.
- A compliance review requiring authorized attention.

Reports will summarize application status, pendency, overdue tasks, compliance indicators, expiry risks, and processing trends. Notification delivery, escalation actions, and subsequent user responses will be recorded for auditability.

4. Innovation Module A: Compliance Evidence Intelligence (CEI)

4.1 Objective

CEI will extend conventional compliance tracking by connecting each applicable regulatory requirement to its supporting evidence and providing an explainable assessment of the evidence's recorded status.

Instead of simply showing that an application is pending, the module will help users understand **which requirement needs attention, what evidence is available, what issue has been identified, and what action is required next.**

4.2 How CEI Will Work

Step 1 — Maintain a requirement register

Store configured requirements with their source references, applicability conditions, expected evidence, responsible role, relevant deadlines, and verification requirements.

Step 2 — Link evidence to requirements

Each uploaded document or evidence record will be associated with the relevant application, institution, document type, and applicable requirement.

Step 3 — Validate evidence metadata

The system will check for configured conditions such as:

- Required evidence not submitted.
- Missing or invalid metadata.
- Recorded expiry dates that have passed or are approaching.
- Mismatched identifiers between related records.
- Duplicate document files or versions.
- Evidence awaiting human verification.

OCR may assist with extracting dates, identifiers, or document text. Extracted values will be treated as provisional until verified where necessary.

Step 4 — Generate an explainable compliance status

The system will assign an evidence status, such as:

- Not submitted.
- Submitted and awaiting verification.
- Verified.
- Action required.
- Expired.
- Rejected by verifier.
- Not applicable.

These statuses describe the evidence and review process. They do not independently establish statutory compliance or approval eligibility.

Step 5 — Present an action-oriented result

For every flagged requirement, the user will be able to see:

- The relevant requirement and its source.
- The evidence submitted or missing.

- The validation result and reason for the flag.
- The current verification status.
- The responsible role and next permitted action.

4.3 Example

Suppose an application requires a particular supporting certificate.

The institution uploads a certificate, but its recorded expiry date has passed.

CEI flags the evidence as potentially invalid or expired, displays the relevant date and requirement, and prompts the institution or authorized officer to take the appropriate action. The authorized verifier reviews the evidence and records the outcome.

The workflow engine can then route the required follow-up task, subject to the configured procedure.

4.4 Expected Benefit

CEI is intended to reduce manual effort in identifying missing or problematic evidence, improve traceability, and make compliance-related follow-up more precise.

Its distinguishing focus is the relationship between **requirements, evidence, validation results, and explainable actions**, rather than document storage or checklist tracking alone.

5. Innovation Module C: Regulatory Change Impact Engine (RCIE)

5.1 Objective

RCIE will help authorized administrators understand the potential operational consequences of changes to regulatory requirements, guidelines, or configured workflow rules.

When an approved or proposed rule changes, the engine will identify which requirements, evidence records, applications, and pending tasks may be affected.

Its central question is:

“If this regulatory requirement changes, which existing cases may need review, and why?”

5.2 How RCIE Will Work

Step 1 — Maintain a regulatory source register

Store relevant official documents and rule versions with their issuing authority, source URL, publication date, effective date, applicable category, and relevant clause or page reference.

Step 2 — Compare rule versions

Display differences between a previously configured rule and a proposed update. Document extraction or AI-assisted summarization may help identify changes, but extracted interpretations must be reviewed against the official source.

Step 3 — Map rules to system dependencies

Connect each approved rule to the requirements, evidence types, applicability conditions, workflow transitions, deadlines, and notification policies that depend on it.

Step 4 — Analyze applicability and potential impact

Use configured conditions such as application category, application status, relevant dates, effective dates, and transitional provisions to identify potentially affected cases.

The results will distinguish:

- Potentially affected applications.
- Applications requiring further review.
- Applications apparently unaffected under the configured rules.
- Cases where available information is insufficient to determine applicability.

Step 5 — Generate an impact report

For each potentially affected case, explain the relevant rule change, the associated requirement or workflow dependency, the reason the case was identified, and the review action suggested.

Step 6 — Require authorized review before activation

An authorized administrator will review the proposed rule configuration and impact analysis before approving changes to operational rules. Decisions about retroactive applicability, exceptions, and transitional provisions will not be inferred automatically.

5.3 Example

Consider a **hypothetical demonstration scenario**, not a claim about an actual AICTE amendment.

An authorized administrator proposes adding a supporting-document requirement for a particular application category, with a specified effective date.

RCIE identifies applications matching the configured category and applicability conditions. It flags cases that may need review, explains why each was identified, and produces an impact report for the authorized administrator.

After the rule interpretation and effective-date treatment are approved, the updated configuration can be made available to CEI and the workflow engine. Affected applications can then be assessed and routed for appropriate follow-up.

5.4 Expected Benefit

RCIE is intended to reduce the manual effort required to trace regulatory changes across operational processes, improve change traceability, and help administrators identify cases that may require reassessment.

Its distinguishing focus is the traceable relationship between **regulatory rule versions, system dependencies, applicability conditions, and affected applications**.

6. How A and C Work Together

CEI and RCIE will operate as complementary modules within the same platform.

1. **RCIE maintains the approved rule configuration.** It records the source, version, applicability, and authorized activation of relevant regulatory requirements.
2. **CEI applies the approved requirements.** It connects applicable requirements to evidence records and identifies missing, expired, inconsistent, or unverified evidence.

3. **The workflow engine coordinates follow-up.** It routes permitted tasks to the responsible roles, tracks deadlines, sends notifications, and records actions.
4. **The dashboard provides visibility.** It displays the application status, evidence issues, outstanding tasks, and potential impact of rule changes.
5. **Authorized personnel retain control.** Regulatory interpretation, evidence verification, and statutory decisions remain subject to the applicable human approval process.

This creates a traceable path from a regulatory source to a configured requirement, supporting evidence, identified issue, and resulting administrative action.

7. User Roles and Dashboards

The platform will use role-based access control, with permissions assigned according to organizational responsibilities.

Institution / Applicant

- View and track their own applications.
- Upload or update supporting documents.
- Review outstanding requirements and compliance alerts.
- Respond to requests for additional information.
- Track relevant deadlines and application progress.

AICTE Administrative / Regional Officer

- View permitted application queues and pendency summaries.
- Monitor outstanding tasks, deadlines, and escalations.
- Review evidence-related flags and follow-up actions.
- Access authorized reports and processing histories.

Scrutiny Committee / EVC Member

- Access assigned cases and relevant documents.
- Record scrutiny observations or EVC-related information.

- Submit reports and track outstanding assigned actions.

Compliance Reviewer

- Review evidence mapped to requirements.
- Verify extracted metadata and validation flags.
- Record verification outcomes and reasons.
- Maintain an auditable evidence-review history.

Competent Authority / Approving Officer

- Access authorized review materials and recommendations.
- Review application status and required reports.
- Record decisions through the permitted approval process.

System Administrator / Authorized Rule Administrator

- Configure roles, workflow rules, notification policies, and requirement mappings.
- Manage regulatory source versions and proposed rule changes.
- Review RCIE impact reports and activate approved configurations.
- Maintain audit and system-configuration records.

These roles are a proposed logical access model; the final role definitions and permissions must match the authorized organizational structure and statutory procedures.

8. Technical Architecture and Technology Stack

The system will use a modular web architecture suitable for a hackathon prototype and capable of being extended after validation.

Frontend

- React with TypeScript.
- Vite for the development and build environment.
- Tailwind CSS and shadcn/ui for the interface.

- React Flow for workflow visualization.
- Recharts for dashboard analytics.

Backend

- Node.js with Express and TypeScript.
- REST APIs for applications, workflow tasks, evidence records, regulatory rules, notifications, and reporting.
- Server-side validation and authorization for all sensitive operations.

Database and storage

- PostgreSQL, potentially managed through Supabase.
- Relational records for applications, requirements, evidence metadata, tasks, rule versions, and audit events.
- Private object storage for uploaded documents, with access controls and authorized download mechanisms.

Authentication and authorization

- Supabase Auth or another suitable identity provider.
- Server-side verification of authentication tokens.
- Role- and permission-based access checks.
- Restricted access to application records and supporting documents.

Background processing

- A persistent job queue or database-backed worker for scheduled reminders, expiry checks, and escalation processing.
- Retry handling and logging for failed notification jobs.

Document intelligence

- Optional Tesseract OCR or a suitable existing OCR service for extracting selected fields from uploaded documents.
- Deterministic validation rules for dates, identifiers, required fields, and document metadata.

- Optional AI-assisted comparison or summarization for regulatory text, with source references and human review.

The initial implementation will prioritize deterministic rules and auditable processing over unnecessary machine-learning complexity. AI will be used only where it provides a clear benefit.

9. Security, Governance and Human Oversight

Because the platform concerns regulatory applications and supporting documents, security and accountability will be built into the design.

The proposed controls include:

- Authentication and role-based authorization.
- Least-privilege access to applications and documents.
- Private document storage and controlled access links.
- Validation of uploaded file types and sizes.
- Protection of sensitive credentials and secrets.
- Audit logs for status changes, evidence verification, rule configuration changes, and administrative actions.
- Version history for evidence and regulatory configurations.
- Human approval for consequential regulatory interpretations and statutory decisions.
- Source references and explanations for automated flags and impact reports.
- Error handling, backups, and appropriate retention policies.

AI-generated interpretations will not automatically become operative regulatory rules. Any operational rule change will require review and authorization under the configured governance process.

10. Implementation Plan

The development will proceed in phases to ensure that the core system works before advanced features are added.

Phase 1 — Core platform

- Authentication and role-based access.
- Application records and status tracking.
- Configurable workflow stages and task assignment.
- Basic dashboards, notifications, and audit history.

Phase 2 — Compliance Evidence Intelligence

- Requirement register and evidence-to-requirement mapping.
- Document uploads and metadata management.
- Expiry monitoring and basic validation rules.
- Verification workflow and explainable action panel.

Phase 3 — Regulatory Change Impact Engine

- Regulatory source register and version history.
- Comparison of two versions of a small, controlled rule set.
- Requirement-to-workflow dependency mapping.
- Identification of potentially affected synthetic applications.
- Impact report and authorized rule-activation process.

Phase 4 — Integration and refinement

- Background reminder and escalation jobs.
- Optional OCR-assisted metadata extraction.
- Combined dashboards and analytics.
- End-to-end testing, access-control checks, and demonstration preparation.

For the hackathon prototype, a small set of verified requirements and synthetic application records will be sufficient to demonstrate the concept. Any hypothetical regulatory amendment used in the demonstration will be clearly labeled as hypothetical.

11. Evaluation and Demonstration

The prototype will demonstrate a complete scenario rather than merely showing separate screens.

A sample institution submits an application and supporting evidence. The workflow engine assigns the required review task. CEI identifies a missing or expired document, explains the issue, and generates a follow-up action. The responsible user responds and an authorized reviewer records the verification outcome.

Next, an administrator proposes a clearly labeled hypothetical regulatory change. RCIE compares the rule versions, identifies potentially affected applications, explains the applicability conditions, and produces an impact report. After authorized review, the approved configuration becomes available to the compliance and workflow modules.

The demonstration will measure:

- Whether missing or expired evidence is identified correctly against the configured rules.
- Whether every compliance flag has an understandable explanation.
- Whether notifications and escalation jobs execute according to their configuration.
- Whether the impact engine identifies the expected cases in a controlled test dataset.
- Whether users can access only the records and actions permitted by their roles.
- Whether rule changes and verification actions can be traced through the audit history.

Results will be measured using test data rather than unsupported claims about production performance.

12. Expected Outcomes

The proposed solution aims to:

- Improve visibility into application status and pending actions.

- Reduce manual effort in tracking compliance evidence and document expiry.
- Provide more precise, explainable compliance follow-up.
- Help administrators assess the potential effects of regulatory changes.
- Improve traceability between regulatory requirements, evidence, and administrative actions.
- Support consistent notification, escalation, and reporting processes.
- Preserve authorized human oversight of regulatory interpretation and statutory decisions.

Actual improvements will need to be established through testing and evaluation.

13. Innovation Summary

The solution combines the required approval workflow and monitoring capabilities with two focused innovation modules:

Compliance Evidence Intelligence (CEI) connects requirements to evidence and explains which compliance-related items need attention.

Regulatory Change Impact Engine (RCIE) connects regulatory rule changes to system dependencies and identifies applications that may need reassessment.

Together, they extend the system beyond basic application tracking by linking regulatory requirements, supporting evidence, potential change impacts, and actionable administrative tasks.

The proposed innovation is the integrated, explainable and traceable approach to compliance evidence management and regulatory change impact analysis. Its originality relative to existing AICTE systems and comparable platforms must be validated before making claims of uniqueness.

Conclusion

The proposed system will provide a unified foundation for approval workflow automation, pendency monitoring, EVC coordination, compliance tracking, document expiry alerts, notifications, escalation management, reporting, and secure role-based access.

Its two selected innovation pillars—Compliance Evidence Intelligence and the Regulatory Change Impact Engine—will add evidence-level compliance insights and traceable analysis of regulatory changes.

By integrating these capabilities with configurable workflows and authorized human review, the system aims to make approval administration more transparent, manageable, explainable, and adaptable to regulatory change.